Keyhole surgery

Keyhole surgery involves performing operations through tiny incisions rather than large, open cuts. Surgery can also be done via a flexible endoscope—a thin tube inserted through a natural opening, such as the mouth.

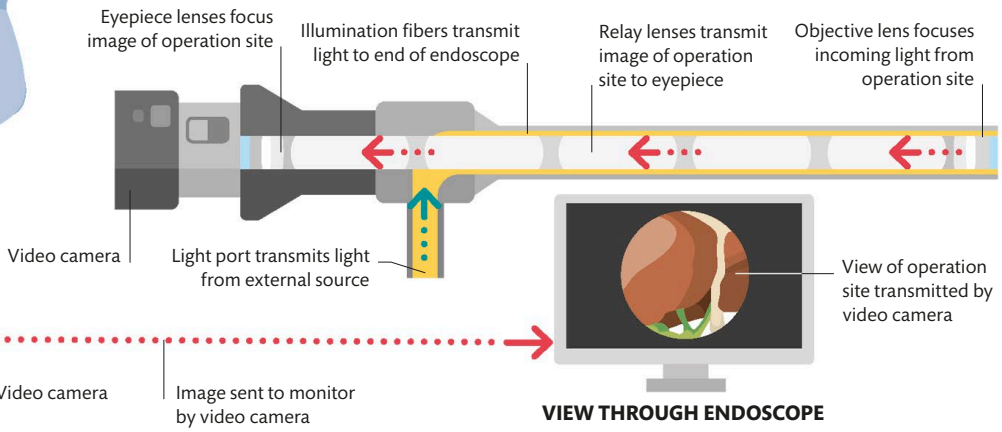
WHEN WAS THE FIRST KEYHOLE SURGERY PERFORMED?

The first keyhole operation was carried out in 1901, on dogs. The first keyhole surgery on humans was performed in 1910.

How keyhole surgery works

Small incisions are made in the skin, and hollow instruments called trocars are inserted into the incisions to keep them open for the endoscope and other instruments. A rigid endoscope transmits light to the operation site. It also enables the surgeon to view the operation site, either directly through an eyepiece or, if a video camera is attached to the eyepiece, on a monitor. Surgical instruments are inserted through separate incisions for tasks such as cutting or stitching tissue or clamping blood vessels.

Rigid endoscope
A rigid endoscope contains fiber-optic cables that transmit light to the operation site and lenses that relay images from the site to an eyepiece. Often, a video camera is attached to the eyepiece, and the image is transmitted to a monitor to provide a clear view for the surgeon.



Keyhole surgery on abdomen
Called laparoscopy, keyhole surgery on the abdomen is performed by using a rigid endoscope. Carbon dioxide gas is pumped into the abdomen to make space around the organs, and the surgeon inserts the laparoscope to provide a view of the operation site. Instruments to carry out the surgery are inserted through other small incisions in the abdomen.

